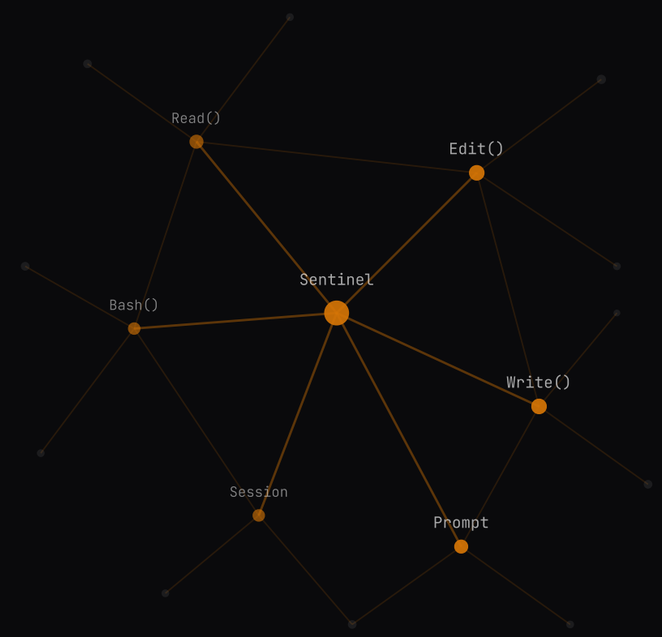
V0.3.0 — "SENTINEL"

How the agent stops re-reading your casebase.

engram v0.3 installs a Claude Code hook layer that fires on every **Read**, **Edit**, **Write**, and **Bash cat**. When the graph has high-confidence coverage of a file, it denies the raw read and delivers a **~300-token structural summary** as the block reason — and your agent sees it as system context, not an error.

```
$ engram install-hook
📌 PreToolUse · PostToolUse · SessionStart · UserPromptSubmit
4 hook events wired · settings.local.json backed up
✅ Installed. The next session intercepts automatically.
```

TOKEN REDUCTION	READ HIT RATE	HOOK HANDLERS	TESTS PASSING
82%	75%	7	439



01 · LIFECYCLE

FOUR HOOK EVENTS · FIRE ORDER DETERMINISTIC

Four hook events.
Seven handlers. One dispatcher.

Claude Code dispatches to each hook via `engram intercept`, piping the raw JSON payload to stdin. Handlers write a JSON response to stdout and exit **0**. Three of the four inject context into your session; the fourth takes notes. Nothing ever blocks the tool call.

HOOK_01 FIRES ON SESSION BOOT

SESSIONSTART

Project brief injection

engram composes a ~500-token brief — graph stats, top god nodes, recent landmines, and the current git branch from `.git/HEAD` — and delivers it to Claude as `additionalContext` before the session sees any tool calls.

Fires on **startup**, **clear**, **compact**. On **resume** it passes through so context isn't double-injected.

→ replaces ~4 exploration reads · -5,000 tok

HOOK_02 FIRES ON EVERY PROMPT

USERPROMPTSUBMIT

Keyword-gated pre-query

Significant terms (≥3 chars, non-stopword) are extracted from the prompt. If there are ≥2 terms and the graph returns ≥3 matches, engram injects the relevant subgraph before Claude begins reasoning.

Privacy invariant: raw prompt text is never written to any log. Only the decision lands in `hook-log.jsonl`.

→ prevents ~1.5 reads per prompt · -8,000 tok

HOOK_03 FIRES ON EVERY TOOL CALL

PRETOOLUSE

Read · Edit · Write · Bash

Read: deny + reason with graph summary. File never actually read. **Edit/Write:** allow + additionalContext with landmine warnings — never blocks. **Bash:** strict parser for single-file `cat/head/tail` delegates to Read.

The *money path*. 60% hit rate on mature projects. Bulk of the savings.

→ replaces full reads · -18,000 tok

HOOK_04 FIRES ON EVERY COMPLETION

POSTTOOLUSE

Pure observer

After the tool runs, the Sentinel logs tool name, file path, output size, success flag, and the PreToolUse decision to `.engram/hook-log.jsonl`. Append-only, atomic, 10 MB rotation.

Never injects. Never modifies tool output. Produces the data v0.3.1 will use for self-tuning.

→ no direct savings · enables auto-tuning

● 02 · THE MONEY PATH NINE CHECKS · ONE DENY

Every Read runs seven gates before the Sentinel decides.

The Read handler is the highest-leverage component. Seven passthrough gates guard against false-positive interceptions. The money path at the bottom is where tokens actually get saved — and it only fires when every gate clears.

src/intercept/handlers/read.ts · handleRead()

GATE 1

Validate payload shape

`tool_name` is Read, `file_path` is a non-empty string, `offset` and `limit` are not set — partial reads can't be satisfied with structural data.

tool_name · file_path · offset · limit

GATE 2

Content safety

Reject 38 binary extensions (.png .pdf .zip .wasm .so ...) and 10 secret patterns (.env, .pem, .key, id_rsa, credentials, secrets.json, .p12, .pfx, .keystore).

binary · secret

GATE 3

Resolve project context

Normalize the path, walk up for `.engram/graph.db`, verify the file lives inside the project root. Ignored zones: `node_modules`, `.git`, `.engram/cache`.

project root · outside · node_modules

GATE 4

Kill switch

If `.engram/hook-disabled` exists, every handler falls through to passthrough. For debugging without uninstalling.

hook-disabled flag

GATE 5

Query graph

Open `.engram/graph.db`, filter nodes where `sourceFile === relPath`, compute `codeNodeCount` excluding file and module metadata.

nodes found · empty graph

GATE 6

Staleness check

If the file mtime is newer than `graph.db`, the graph is stale for this file — passthrough so Claude sees current contents. Refresh with `engram init`.

file newer than graph

GATE 7

Confidence threshold

`confidence = min(codeNodeCount / 3, 1) × avgNodeConfidence`. Must be ≥ 0.7. Conservative — prefer passthrough over a sparse summary that misleads Claude.

≥ 0.7 · < 0.7

MONEY →

DENY · reason = structural summary

Render up to 600 tokens of NODE and EDGE lines sorted by degree, with an escape-hatch footer telling Claude how to Read with explicit offset/limit if full lines are needed. The file is never actually read.

STDOUT · HOOK RESPONSE

EXIT 0


```
{
  "hookSpecificOutput": {
    "hookEventName": "PreToolUse",
    "permissionDecision": "deny",
    "permissionDecisionReason": "[engram] Structural
      summary for src/core.ts

      Nodes: 13 | avg extraction
      confidence: 1.00

      NODE init() [function] L48
      NODE query() [function] L131
      NODE path() [function] L145
      NODE godNodes() [function] L159
      NODE stats() [function] L173
      NODE getFileContext() [function] L270
      NODE mistakes() [function] L377
      NODE benchmark() [function] L418
      NODE learn() [function] L182
      NODE getDbPath() [function] L21

      Key relationships:
      EDGE core.ts --imports--> query.ts
      EDGE core.ts --contains--> init()
      EDGE core.ts --contains--> getFileContext()

      Note: engram replaced a full
      file read with this structural
      view to save tokens. Read with
      explicit offset/limit if you
      need specific lines."
  }
}
```


RAW · 4,169 TOK

SUMMARY · 300 TOK

03 · WHAT IT TOUCHES

SIX SURFACES · ONE OWNED

Six surfaces engram reads. One it owns.

The Sentinel is purely additive. It consumes signals from surfaces you already have — git history, existing hooks, rule files — and writes to exactly one place it owns: the local graph database. Every other tool in your stack runs untouched.

.engram/graph.db

OWNED

The graph

Local SQLite file, mined by engram's **AST miner** (10 languages, zero LLM cost), **git miner** (co-change patterns), and **session miner** (`bug:` / `fix:` lines). One file per project, fully portable. The only thing engram writes outside the marker blocks in your rules files.

CLAUDE.md · .cursorrules · AGENTS.md

WRITE

Your rules files

Hand-written project rules in your repo root are scanned by the session miner for decisions, patterns, and mistakes. `engram gen` writes a summary INTO the file between marker blocks — anything above or below is preserved verbatim.

.git/ · .git/HEAD · git log

READ

Your git history

The git miner reads `git log` for co-change patterns and hot files. The SessionStart hook reads `.git/HEAD` for the current branch name when composing the project brief. Pure reads — no LFS touched, no refs modified.

.claude/settings.json · settings.local.json

WRITE

Claude Code hook config

The installer writes four entries (PreToolUse, PostToolUse, SessionStart, UserPromptSubmit) alongside whatever hooks you already have. **Idempotent. Atomic. Timestamped backup** before any write. Surgical uninstall preserves everything else.

Any MCP server you run

PEER

Other MCP tools

engram ships its own MCP server with six tools (`query_graph`, `god_nodes`, `graph_stats`, `shortest_path`, `benchmark`, `list_mistakes`) but never conflicts with other servers. Run it alone or alongside any stack you have.

Claude Code · Cursor · aider · stdio JSON-RPC

HOST

Your AI client

The Sentinel hook protocol is Claude Code-specific. The CLI (`engram query`, `engram gen`) and MCP server work with anything that speaks stdio JSON-RPC. **One graph, many clients.**

● 04 · MEASUREMENTS

FOUR FILES · REAL DOGFOOD

Not projected. Measured on real engram code.

Four representative files from this repo. Three intercepted, one passthrough (correctly — only one exported function, below the confidence threshold). **82 % token reduction · 75 % hit rate.** Reproducible by anyone with `engram hook-preview <file>`.

$ engram hook-preview · src/*.ts

dogfood

FILE	NODES	RAW READ	SENTINEL	SAVINGS	DECISION
src/core.ts	13	4,169	~300	-3,869	DENY
src/graph/query.ts	10	4,890	~300	-4,590	DENY
src/intercept/dispatch.ts	5	1,820	~300	-1,520	DENY
src/intercept/handlers/read.ts	1	1,310	1,310	0	PASSTHROUGH
totals	**29**	**12,189**	**2,210**	**-9,979**	**-82 %**

$ engram hook-stats

SESSION TOKEN REDUCTION

82 measured

READ HIT RATE

75 measured

HOOK COLD START

131 ms

TEST SUITE

1.5 s · 439 tests

BINARY SIZE

42.5 kb packed · 9 files

05 · QUICK REFERENCE

EIGHT COMMANDS

Eight commands.
The whole surface.

Everything else in the code is a passthrough branch or a safety invariant. These eight are what you actually type.

INSTALL

```
$ engram install-hook
```

Writes four hook entries to `.claude/settings.local.json`. Atomic. Idempotent. Backed up.

DRY-RUN

```
$ engram hook-preview
```

Show what the Read handler *would* do for a file. Deny, allow, or passthrough — with the reason.

MEASURE

```
$ engram hook-stats
```

Summarize `.engram/hook-log.jsonl`. Per-event, per-tool, per-decision. Token savings.

KILL SWITCH

```
$ engram hook-disable
```

Touch `.engram/hook-disabled`. Every handler checks it and passes through. Reverse with `hook-enable`.

REBUILD

```
$ engram init
```

Re-mine the project. ~40 ms, zero LLM calls. Run after large structural changes to refresh confidence.

QUERY

```
$ engram query "…"
```

Direct graph query via BFS with a 2,000-token budget. Use when the hook didn't catch what you needed.

ROLLBACK

```
$ engram uninstall-hook
```

Surgical removal. Preserves every other hook in settings.json. Automatic backup before the write.

UPGRADE

```
$ npm i -g engramx
```

Pull the latest from npm. v0.3.1 will tune confidence thresholds from real hook-log data.

engram

V0.3.0 · SENTINEL

Context as infra for your AI coding tools. Hook-based Read/Edit/Write/Bash interception, zero LLM cost, zero cloud, zero native deps. Published 2026-04-11. Apache-2.0.

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